

PyeLink and SyeLink

Open-source Python tools for low-level EyeLink 1000 plus control and data parsing

Mohammadhossein Salari^{1*} // Marcus Nyström² // Diederick C. Niehorster^{2,3} // Roman Bednarik¹

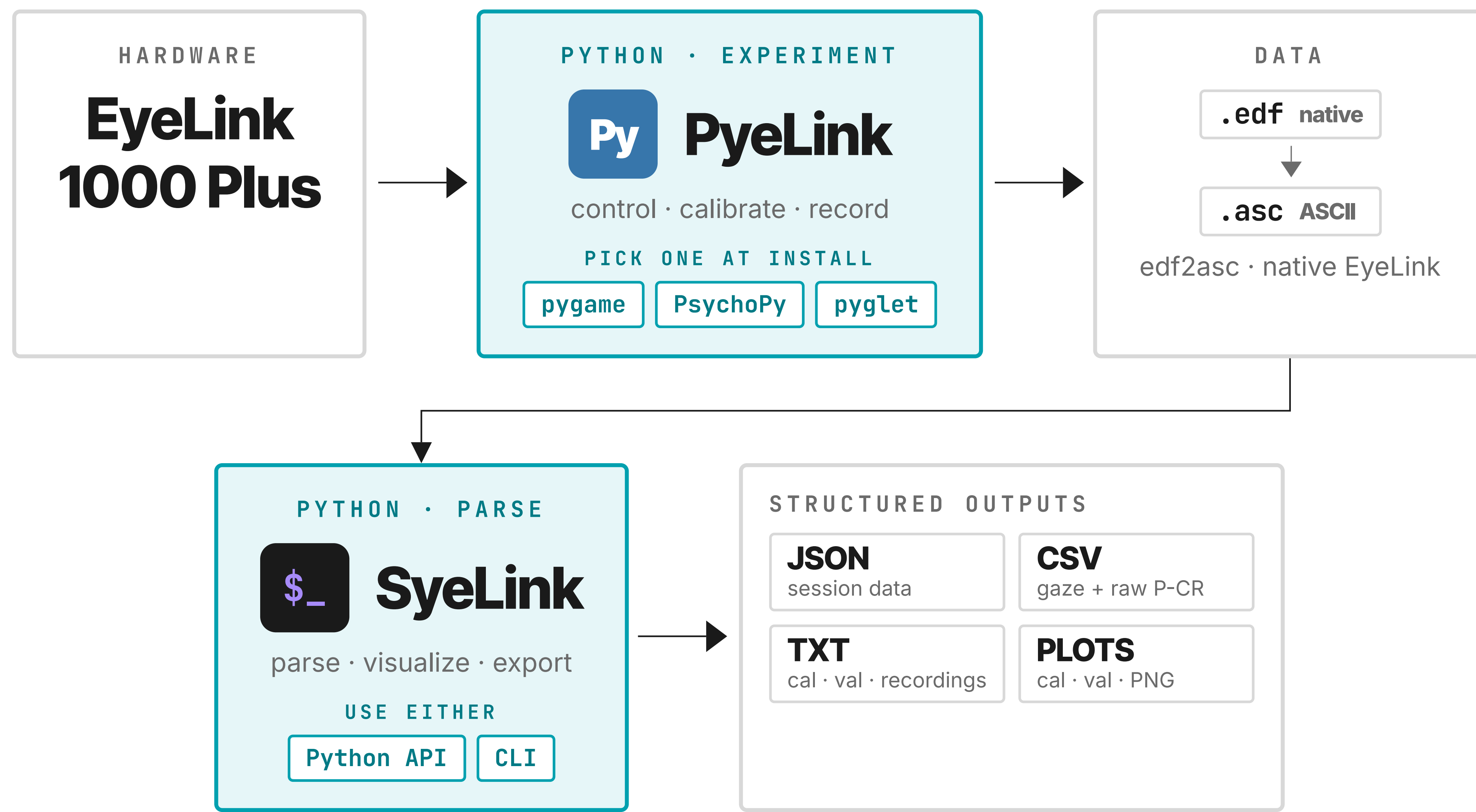
¹School of Computing, University of Eastern Finland, Joensuu, Finland

²Lund University Humanities Lab, Lund University, Lund, Sweden

³Department of Psychology, Lund University, Lund, Sweden

ETRA '26 · Marrakesh, Morocco · June 1–4, 2026 · DOI 10.1145/3797246.3805844

One Python workflow for EyeLink — on any OS!



PyeLink `p[python e]yelink · experiment control`

PyeLink is a cross-platform Python wrapper around SR Research's `pylink` that gives you full control of the EyeLink 1000 Plus. Pick `pygame`, `PsychoPy`, or `pyglet` as the display backend at install, then design, calibrate, and run your experiment entirely in Python — and if you need them, capture raw P-CR position and pupil size, in both camera and HREF coordinates, for the entire duration of the experiment, even during calibration!

```
import pyelink as el

settings = el.Settings(
    backend='pygame', # or 'psychopy', 'pyglet'
    screen_res=(1920, 1080),
    screen_distance_top_bottom=(600.0, 640.0),
    record_raw_pupil_cr=True, # raw P-CR + gaze
    filename='session_01',
)
tracker = el.EyeLink(settings)
tracker.calibrate(record_samples=True)
tracker.start_recording()
# ... show stimuli, collect gaze ...
tracker.stop_recording(); tracker.end_experiment()
```

Py pyelink PyPI MIT

```
pip install pyelink[pygame]
```

docs · pyelink.readthedocs.io
code · github.com/mh-salari/pyelink

SyeLink `s[tructured e]yelink · parse & visualize`

SyeLink reads EyeLink `.asc` files and turns them into structured JSON and CSV files, along with calibration and validation plots. It also handles files that contain raw P-CR alongside calibrated gaze — something existing parsers can't do. You can use it from the command line or as a Python library.

```
$ syelink convert data.asc
✓ data.json           all session data
✓ data_samples.csv    gaze + raw P-CR (32 cols)
✓ calibrations.txt    validations.txt    recordings.txt

$ syelink info data.json
$ syelink plot-validation data.json -i 0 -o val.png
$ syelink export-samples data.asc -o samples.csv
```

\$ syelink PyPI MIT

```
pip install syelink
```

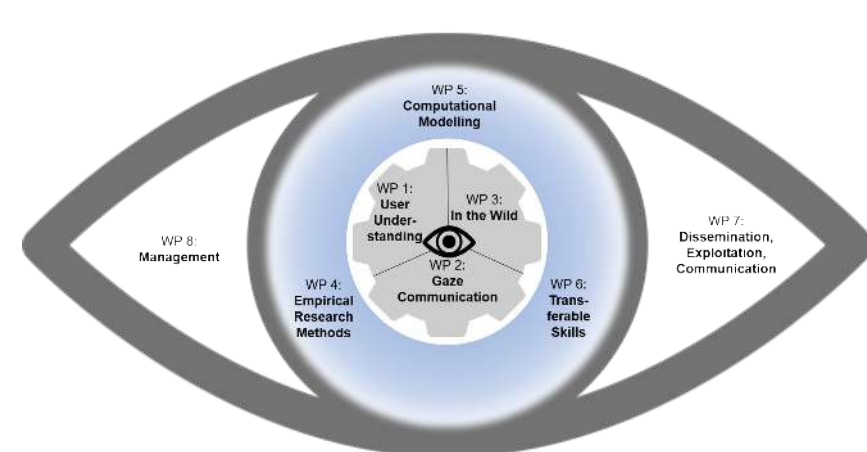
docs · syelink.readthedocs.io
code · github.com/mh-salari/syelink

Acknowledgments

Funded by the EU Horizon Europe programme, grant number 101072410 Eyes4ICU project.



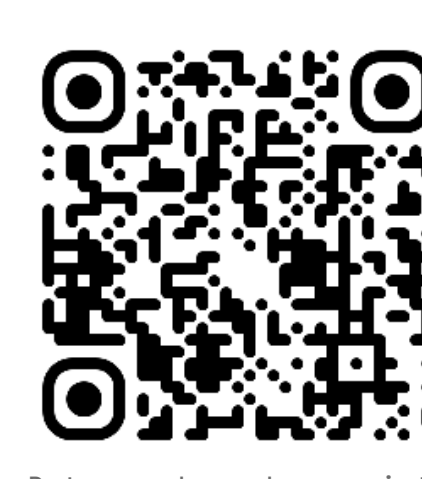
Funded by
the European Union



UNIVERSITY OF
EASTERN FINLAND



LUND
UNIVERSITY



Data, code and more into



Mohammadhossein.Salari@uef.fi